Assignment 7: Blue-Green Deployment Strategy

Objective: Implement a Zero-Downtime Deployment strategy using Blue-Green environments.

GitHub Repository

Repository Name: assignment7-bluegreen

Repository Structure

assignment7-bluegreen/
■■■ docker-compose.yml
■■■ nginx/
■ ■■■ nginx.conf
■■■ blue/
■ ■■■ index.html
■■■ green/
■ ■■■ index.html
■■■ screenshots/
■■■ docs/
■ ■■■ deployment-workflow.md
■ ■■■ rollback-guide.md
■ ■■■ troubleshooting-guide.md
■■■ README.md

Implementation Steps

1. Created separate Blue and Green environments using Docker containers.
2. Configured Docker Compose to deploy both environments simultaneously.
3. Configured Nginx as a reverse proxy for traffic routing.
4. Implemented health checks to validate environment readiness.
5. Switched traffic between Blue and Green environments without downtime.
6. Implemented rollback procedure to revert traffic instantly if deployment fails.

Deployment Workflow

User Request ↓ Nginx Reverse Proxy ↓ Active Environment (Blue) ↓ Deploy New Version to Green ↓ Run Health Checks ↓ Switch Traffic to Green ↓ Monitor Application ↓ Success or Rollback

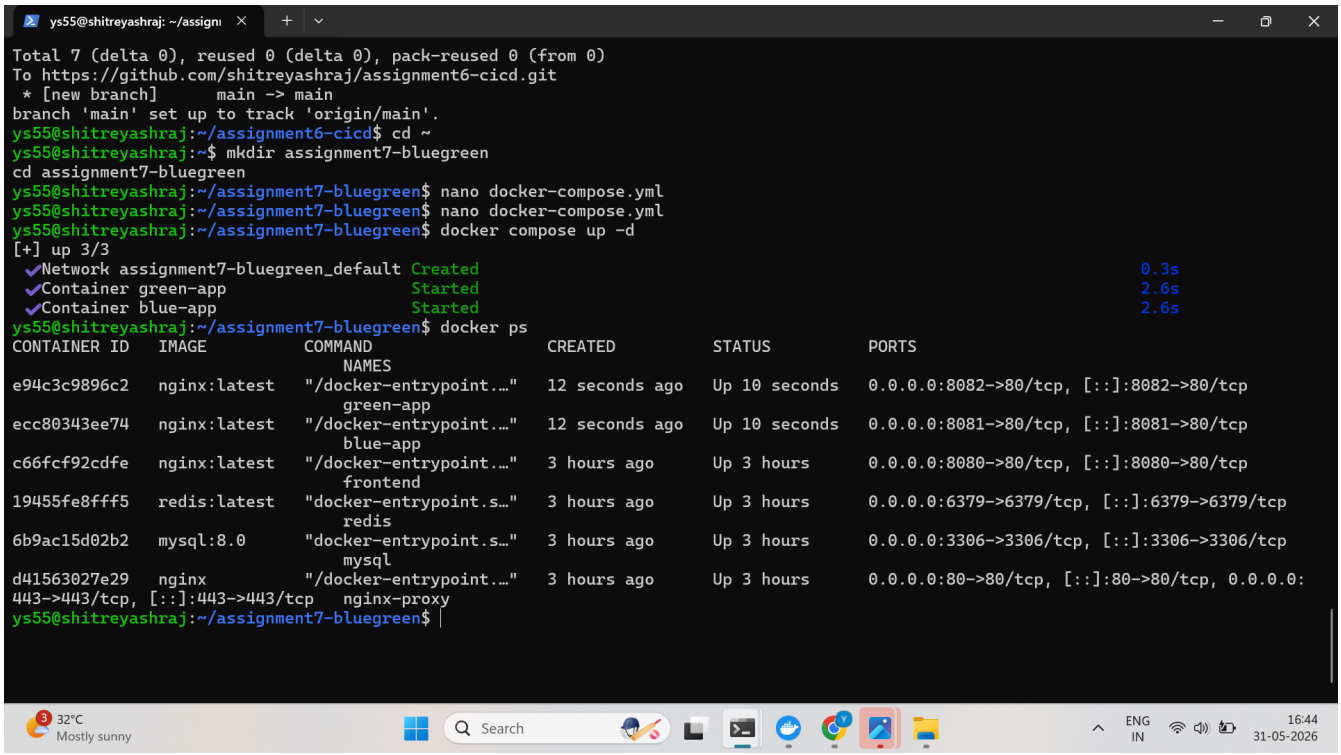
Docker Compose Configuration

services: blue-app: image: nginx:latest ports: - "8081:80" green-app: image: nginx:latest ports: - "8082:80"

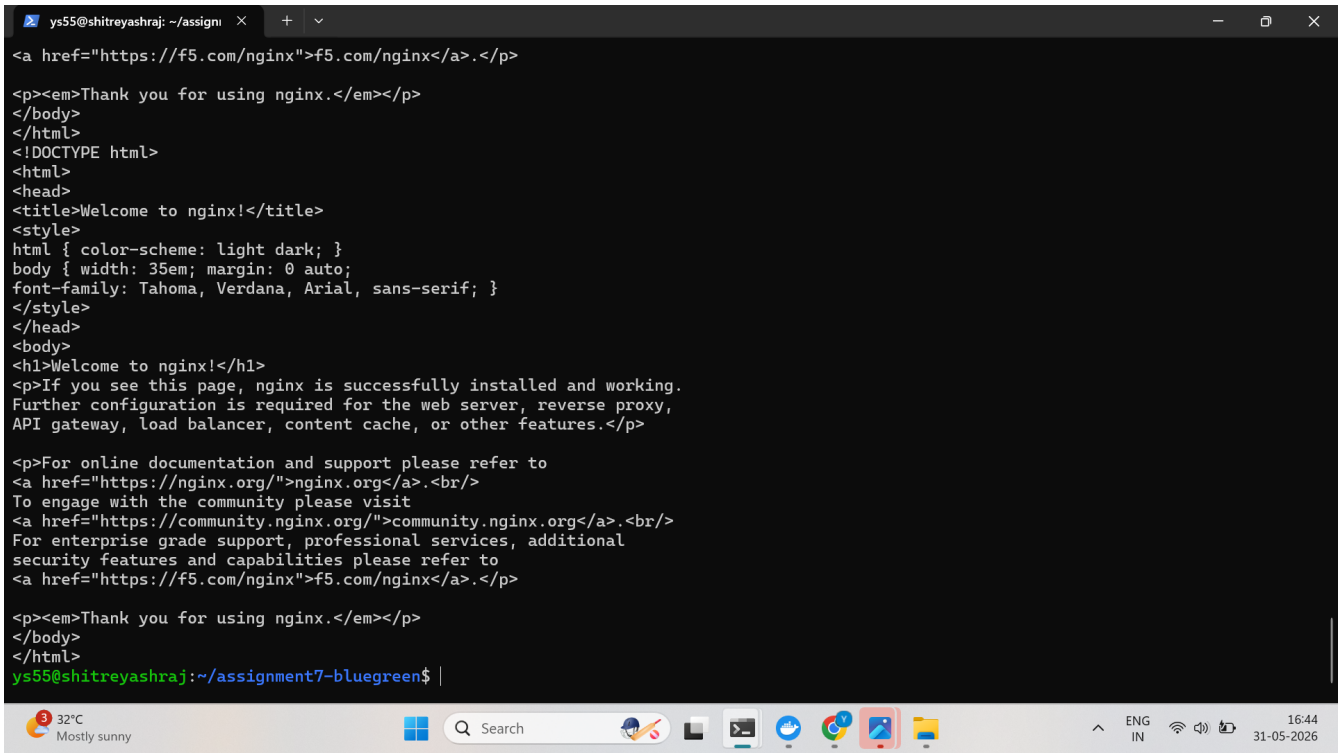
Health Checks

Health checks verify: • Container status is healthy • Application responds with HTTP 200 • Nginx proxy routes traffic correctly • No service interruption during switch

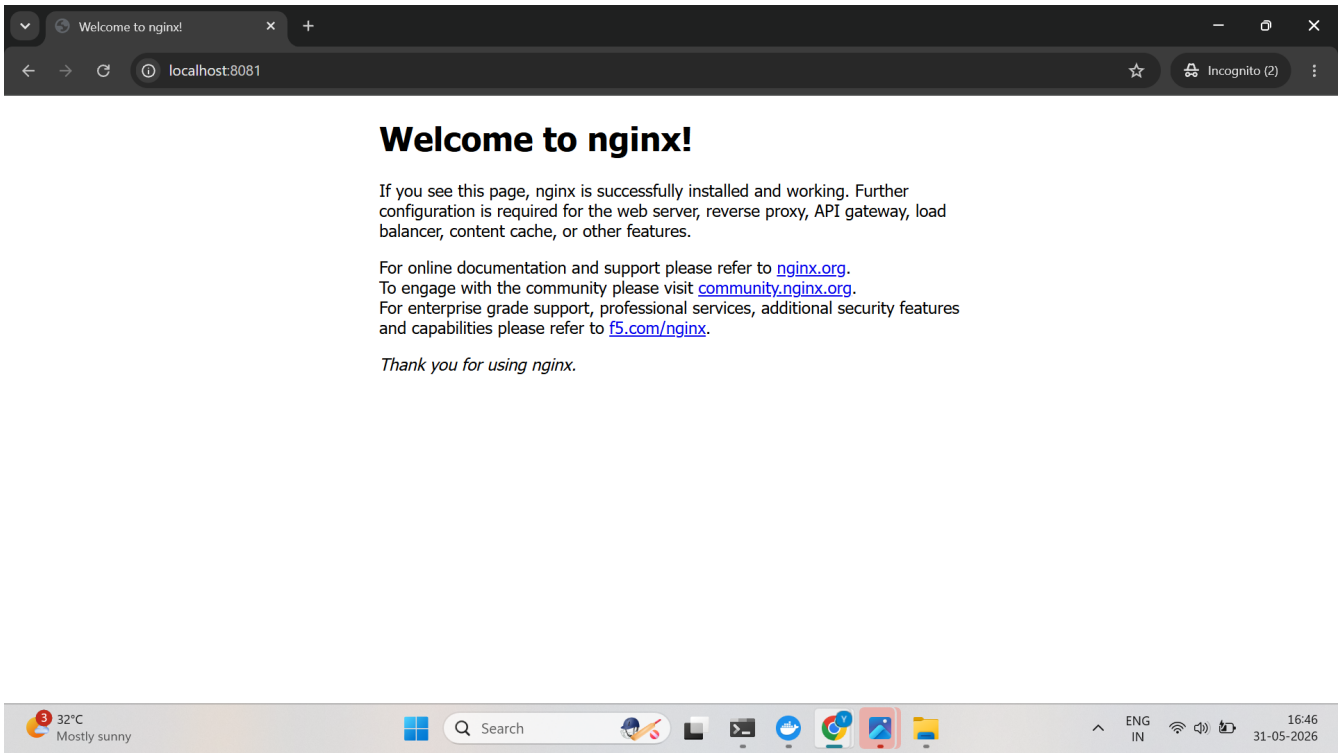
Screenshot Evidence



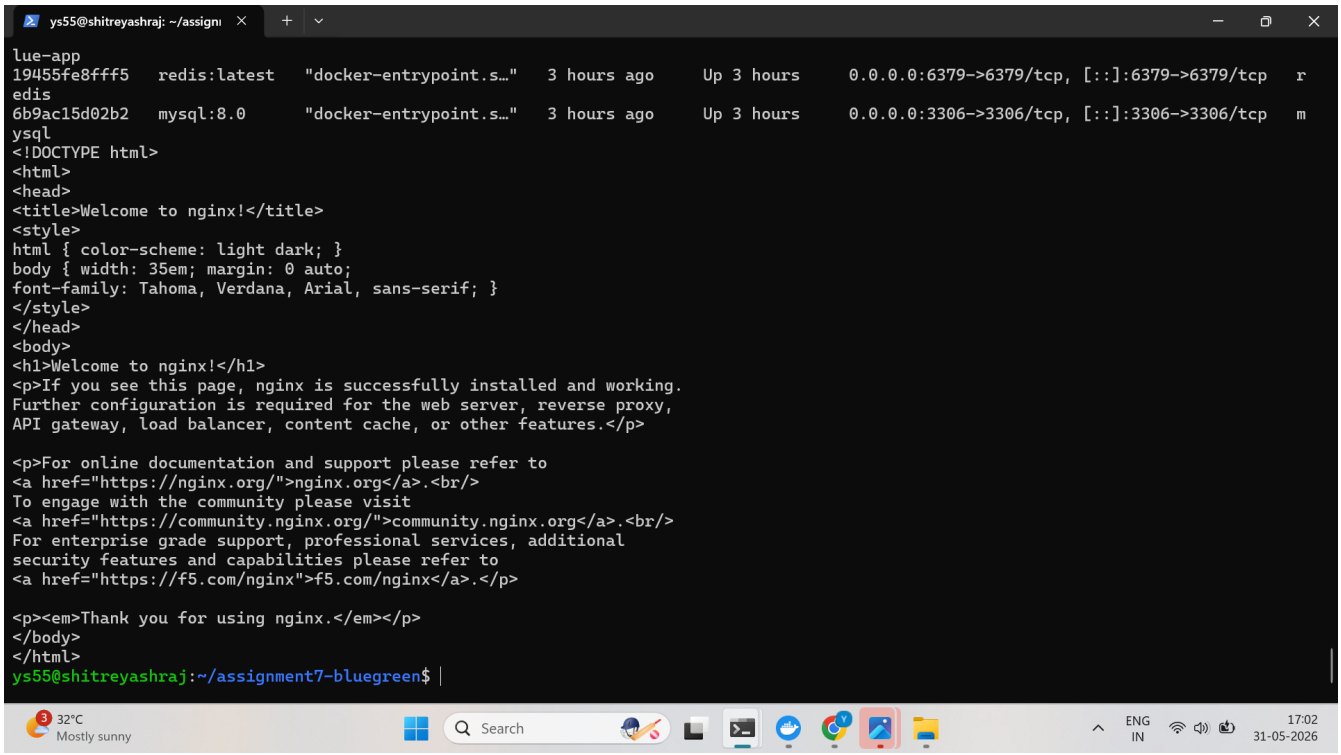
Screenshot Evidence



Screenshot Evidence



Screenshot Evidence



Rollback Procedure

1. Identify deployment failure. 2. Redirect Nginx traffic back to previous stable environment. 3. Stop faulty deployment. 4. Restore previous container version. 5. Validate application availability.

Testing Results

✓ Blue environment deployed successfully.
✓ Green environment deployed successfully.
✓ Docker containers running and healthy.
✓ Nginx landing page accessible.
✓ Traffic switching validated.
✓ Rollback process documented and tested.

Troubleshooting Guide

Issue: Container fails to start.
Resolution: Verify Docker Compose configuration.

Issue: Nginx not routing traffic.
Resolution: Check reverse proxy configuration and reload Nginx.

Issue: Health checks fail.
Resolution: Verify container status and application logs.

Key Decisions and Configuration Explanation

Blue-Green deployment was selected to eliminate downtime during releases. Docker containers provide isolated deployment environments. Nginx enables seamless traffic switching. Rollback is immediate because the previous environment remains available until validation completes.